Functional Annotation Report: *betA* (Choline dehydrogenase, CDH) — *Escherichia coli* K-12

UniProt: P17444 · **Gene:** *betA* (b0311, JW0303) · **EC 1.1.99.1** (choline dehydrogenase); annotated also as **EC 1.2.1.8** (betaine aldehyde dehydrogenase) · **Family:** GMC oxidoreductase superfamily

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0. Gene Identity Verification

Identity CONFIRMED — unambiguous. The gene symbol *betA* (b0311/JW0303) in *E. coli* K-12 corresponds precisely to the UniProt P17444 annotation "Oxygen-dependent choline dehydrogenase." All independent lines agree: (i) the symbol *betA* is used throughout the primary literature for *E. coli* choline dehydrogenase (P 3065456 P 8626294); P 39510372 (ii) the organism is *E. coli* K-12; (iii) the GMC oxidoreductase family and the FAD-binding/GMC domain architecture (IPR011533, IPR012132, IPR000172, IPR007867, IPR036188) match the characterized flavoenzyme ( ); P 39510372 and (iv) no competing gene with the same symbol/organism was encountered. There is a well-populated, specific literature for this exact protein, so research proceeded on the correct target.

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1. Summary / Answer to the Research Question

betA encodes **choline dehydrogenase (CDH/CHDH)**, a **membrane-associated, FAD-dependent flavoenzyme** of the glucose–methanol–choline (GMC) oxidoreductase superfamily. Its primary function is to catalyze the **first, committed oxidation step of glycine betaine biosynthesis**: the oxidation of **choline to betaine aldehyde** (EC 1.1.99.1). Betaine aldehyde is then oxidized to the osmoprotectant **glycine betaine**, a reaction carried out mainly by the dedicated enzyme **betaine aldehyde dehydrogenase (BetB)**, although BetA itself shows a weaker intrinsic capacity to complete this second step (the basis of its secondary EC 1.2.1.8 annotation). The enzyme operates on the **cytoplasmic face of the inner (plasma) membrane**, is a **peripheral membrane protein** that associates as a **trimer** in a native lipid environment, and transfers reducing equivalents to the **membrane respiratory chain** (hence its strict requirement for oxygen/aerobic electron acceptors). The pathway allows *E. coli* to convert exogenous choline into glycine betaine, a potent **compatible solute** accumulated to resist **hyperosmotic stress**.

Gene identity is confirmed: the symbol *betA*, the organism (*E. coli* K-12), the GMC oxidoreductase family, and the BetA/GMC domain architecture (IPR011533 BetA; IPR012132 GMC oxidoreductase; IPR000172/IPR007867 GMC N- and C-terminal domains; IPR036188 FAD/NAD-binding fold) are all mutually consistent with the choline-oxidizing flavoprotein described in the literature below.

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2. Primary Molecular Function

2.1 Reaction catalyzed and substrate specificity

Choline dehydrogenase catalyzes:

> **Choline + acceptor → betaine aldehyde (+ reduced acceptor)** (EC 1.1.99.1)

A recent purification and kinetic study of the *E. coli* enzyme established that CHDH **"is a membrane-bound enzyme belonging to the glucose-methanol-choline (GMC) oxidoreductase superfamily, which is characterized by a crucial FAD-binding domain essential for catalytic function,"** and that **"CHDH catalyzes the oxidation of choline to betaine aldehyde"** (Ma *et al.*, 2024,

).

P 39510372

The same study showed that **"highly pure *E. coli* CHDH could only oxidize choline to betaine aldehyde but failed to further oxidize betaine aldehyde to betaine"**, and that a catalytic **histidine** residue is crucial for activity. This pins the enzyme's **physiologically dominant substrate as choline**, with betaine aldehyde as the product.

Genetic evidence corroborates the assignment: disruption of *betA* **"abolished the ability of *Ser. entomophila* to utilize choline as a carbon source"** in a close relative carrying the *E. coli*-like *betTIBA* genes (Sheen *et al.*, 2013,

),

P 23110363

and in *Sinorhizobium meliloti* a CDH-deficient mutant **"failed to oxidize [methyl-14C]choline to [methyl-14C]betaine"** (Pocard *et al.*, 1997,

).

P 9141699

2.2 Cofactor

BetA is a **flavoprotein using FAD** as the redox cofactor (FAD-binding domain essential for catalysis; Ma *et al.*, 2024,

).

P 39510372

The "99" in EC 1.1.99.1 signifies that the physiological electron acceptor is **not NAD(P)** but a membrane/respiratory acceptor — consistent with its membrane localization and oxygen dependence (see §3, §4).

2.3 Apparent bifunctionality (secondary EC 1.2.1.8 annotation)

UniProt annotates BetA with a second activity, **betaine aldehyde dehydrogenase (BADH, EC 1.2.1.8)**. This reflects reports that the enzyme can, under some conditions, oxidize betaine aldehyde onward to glycine betaine: - In transgenic tobacco, **"choline dehydrogenase (CDH), encoded by *betA*, oxidizes choline to betaine aldehyde which is further oxidized to glycine betaine by the same enzyme,"** and CDH alone sufficed to accumulate glycine betaine (Holmström *et al.*, 2000,

).

P 10938824

- In *Halomonas elongata*, **"BetA is able to oxidize both choline and glycine betaine aldehyde**

and therefore can mediate both steps in the synthesis of glycine betaine" (Cánovas *et al.*, 2000,

). P 10708384

However, the most rigorous recent work on the **purified *E. coli*** enzyme found it does **not** efficiently perform the second oxidation (Ma *et al.*, 2024,

). P 39510372

The best-supported interpretation is therefore: **BetA's primary role is choline → betaine aldehyde**, while the second oxidation (betaine aldehyde → glycine betaine) is principally the job of the dedicated **BetB** aldehyde dehydrogenase; BetA retains only a weak, context-dependent capacity for that step.

2.4 Catalytic mechanism (inferred from GMC-superfamily homologs)

BetA's detailed mechanism has not been solved at atomic resolution, but it is strongly constrained by its closest mechanistically characterized relative, **choline oxidase from *Arthrobacter globiformis*** (also a GMC-superfamily flavoprotein that oxidizes choline → glycine betaine). In that system: - The alcohol substrate is first **activated to an alkoxide** by removal of the hydroxyl proton by an active-site base, then **"alcohol oxidation occurs … through quantum mechanical transfer of a hydride ion from the alpha-carbon of the chelated, alkoxide species to the N(5) atom of the enzyme-bound flavin"** (Gadda, 2008,

). P 19053234

- A **fully conserved active-site histidine (His466)** is required for alcohol oxidation (but not O_2 reduction); its mutation lowered k_cat ~60-fold and k_cat/K_m ~1000-fold (Ghanem & Gadda, 2005,

). P 15654745

An essential catalytic **His** is likewise reported for *E. coli* CHDH (Ma *et al.*, 2024,

). P 39510372

- For the second (aldehyde) half-reaction, **betaine aldehyde exists ~99% as the gem-diol (hydrate) in water**, and that hydrated species is the one oxidized, with a catalytic base of pK_a ~6.7 (Fan *et al.*, 2006,

). P 16460045

Key mechanistic distinction from choline oxidase: BetA is a **dehydrogenase** (EC 1.1.99.1) that reoxidizes its reduced FAD by passing electrons to **membrane respiratory electron carriers**, whereas *A. globiformis* choline oxidase (EC 1.1.3.17) uses **molecular O_2** as the terminal acceptor, producing H_2O_2. This is why BetA is membrane-localized, requires aerobic respiration, and is shut off anaerobically by ArcA (see §3–4) rather than acting as a soluble H_2O_2-generating oxidase.

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3. Subcellular Localization and Quaternary Structure

- **Membrane association:** CHDH is **membrane-bound** but is a **peripheral membrane protein that lacks transmembrane helices**; styrene-maleic-acid (SMA) copolymer, which "can preserve the natural bio-membrane environment," was effective "in extracting peripheral membrane proteins like CHDH" (Ma *et al.*, 2024,). P 39510372 It is located on the **cytoplasmic side of the inner membrane**, the compartment where cytoplasmic choline is oxidized and where electrons feed into the respiratory chain.
- **Oligomeric state:** The enzyme is a **trimer** in a native-like lipid environment (SMA nanodiscs) but appears **monomeric** in the detergent DDM, and SMA-embedded enzyme **"displayed greater catalytic activity"** than DDM-solubilized enzyme — indicating that the membrane/lipid environment supports the active oligomer (Ma *et al.*, 2024,). P 39510372

3b. Structural and Evolutionary Architecture (inference from domains/homology)

BetA has the **canonical bilobal GMC-oxidoreductase fold**, as read from its InterPro domain architecture: - **FAD/NAD(P)-binding structural superfamily fold** (IPR036188) + **GMC oxidoreductase N-terminal FAD-binding domain** (IPR000172): an N-terminal **Rossmann-type (βαβ / ADP-binding) lobe** that binds the FAD cofactor. The purified enzyme's "crucial FAD-binding domain [is] essential for catalytic function"

( ). P 39510372

- **GMC oxidoreductase C-terminal domain** (IPR007867): the **substrate-binding/catalytic lobe** that binds choline and positions its α-carbon against the flavin N5 for hydride transfer. - **GMC oxidoreductase signature** (IPR012132) and a **BetA-specific signature** (IPR011533) classify it within the family.

Evolutionary logic: GMC-superfamily enzymes (glucose oxidase, methanol oxidase, cholesterol oxidase, choline oxidase, choline dehydrogenase) **"exhibit little sequence similarity in their substrate binding domains, but share a highly conserved catalytic site, suggesting a similar activation mechanism for the oxidation of their substrates"** ( ). P 15654745

This explains why *E. coli* CHDH and *A. globiformis* choline oxidase share a conserved catalytic **His** and the same hydride-transfer chemistry despite low global identity (e.g., CDH shares only **~21% identity** with *A. globiformis* choline oxidase;). P 9141699

Functional identity in this family is carried by a small conserved catalytic core rather than overall sequence, consistent with divergent evolution from a common flavin-dependent alcohol-oxidase ancestor.

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4. Biochemical Pathway and Physiological Context

4.1 The glycine betaine biosynthesis pathway (choline → glycine betaine)

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            BetT                 BetA (CDH)              BetB (BADH) / [BetA, weak]
choline(out) ──► choline(in) ──────────► betaine aldehyde ──────────────────────────► glycine bet
           transport        [FAD; O2/respiratory          [NAD+-dependent]
                             electron acceptor]
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- **BetT** — high-affinity, **proton-driven** choline transporter of the **BCCT (betaine-choline-carnitine transporter) family** with 12 transmembrane spans and cytoplasmic N/C termini; it is osmotically activated (~10-fold increase in choline transport after an NaCl upshock) (Tøndervik & Strøm, 2007,

; P 17322201 Ziegler *et al.*, 2010,). P 20923416 BetT imports the choline substrate that BetA oxidizes.

- **BetA** — choline dehydrogenase (this gene): choline → betaine aldehyde.
- **BetB** — betaine aldehyde dehydrogenase: betaine aldehyde → glycine betaine (NAD^+-dependent).

Glycine betaine is a **compatible solute (osmoprotectant)** and methyl donor that *E. coli* accumulates to maintain turgor and protein stability under high osmolarity. The original cloning study mapped and named the structural genes, assigning **"the *betA* gene for the choline dehydrogenase, the *betB* gene for the betaine aldehyde dehydrogenase, and the *betT* gene or operon for the high-affinity choline transport,"** with gene order *betA,B,T* (Andresen *et al.*, 1988,). P 3065456

4.1a Why the product matters: glycine betaine as a compatible solute

The physiological payoff of BetA's reaction is the production of **glycine betaine, one of the most effective bacterial compatible solutes (osmoprotectants)**. Compatible solutes **"can make important contributions to the restoration of the turgor under conditions of low water activity by counteracting the efflux of water from the cell"** and exert **"a stabilizing... effect on the native structure of proteins and cell components"** (review,); P 18062278 glycine betaine specifically stabilizes the native conformation of enzymes under osmotic and chill stress ( ). P 16645311 Expressing the *E. coli bet* region (including *betA*) confers **increased osmotic tolerance in the presence of choline**, and this tolerance was transferable to *Salmonella* ( ). P 3065456 Importantly, **choline itself is not osmoprotective** — it must be oxidized by BetA (and BetB) to glycine betaine — so BetA is the key biosynthetic conversion that unlocks osmoprotection. In enteric *E. coli*, glycine betaine is accumulated as an end-product osmolyte rather than

catabolized (contrast with pseudomonads, where choline/betaine can additionally serve catabolism;

). P 18156257

4.2 Regulation (operon context)

The *bet* system is the **betI–betT / betIBA** regulon with divergent promoters: **"It comprises a regulatory gene, *betI*, and three structural genes: *betT* (choline porter), *betA* (choline dehydrogenase), and *betB* (betaine aldehyde dehydrogenase). The bet genes are regulated by oxygen, choline, and osmotic stress"** (Lamark *et al.*, 1996,). P 8626294

Key control inputs: - **BetI** is a **repressor** relieved by the inducer **choline**. - **Osmotic stress** induces the promoters (BetI-independent, H-NS-independent). - **ArcA represses the promoters anaerobically** — a logical control, because the enzyme requires a respiratory (oxygen-linked) electron acceptor and is non-functional under anaerobiosis.

This regulatory logic reinforces the mechanistic picture: BetA's catalysis is **aerobic/respiration-coupled**, and the pathway is switched on precisely when choline is available and cells face osmotic challenge.

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5. Evidence Hierarchy

Claim	Evidence type	Reference
CHDH is membrane-bound GMC-family FAD flavoenzyme; oxidizes choline → betaine aldehyde; catalytic His; trimer in native lipid	**Direct biochemical / enzymology on purified *E. coli* enzyme**	Ma *et al.* 2024 (P 39510372)
betA = choline dehydrogenase; operon structure & map; osmotic regulation	Molecular genetics / cloning (*E. coli*)	Andresen *et al.* 1988 (P 3065456)
Regulation by choline (BetI), osmotic stress, oxygen (ArcA)	Promoter/reporter genetics (*E. coli*)	Lamark *et al.* 1996 (P 8626294)
BetA necessary for choline catabolism/oxidation	Mutant/genetic (relatives)	Sheen 2013 (P 23110363); Pocard 1997 (P 9141699)
BetA can (weakly) complete both oxidation steps	Heterologous expression	Holmström 2000 (P 10938824); Cánovas 2000 (P 10708384)
BetT = BCCT-family choline transporter feeding the pathway	Review / structural	Ziegler 2010 (P 20923416)
Catalytic mechanism: alkoxide formation + hydride transfer to FAD N5; conserved catalytic His; betaine aldehyde oxidized as gem-diol	**Mechanistic enzymology on GMC homolog (choline oxidase)**	Gadda 2008 (P 19053234); Ghanem 2005 (P 15654745); Fan 2006 (P 16460045)

Claim	Evidence type	Reference
BetT: 12-TM proton-driven, osmotically activated (10-fold)	Membrane topology / transport (*E. coli*)	Tøndervik 2007 (PMID 17322201)
Bilobal GMC fold: N-terminal FAD/Rossmann lobe + C-terminal substrate lobe; conserved catalytic core despite low global identity	**Domain architecture / comparative sequence**	InterPro (IPR000172/007867/012132/011533); PMID 15654745 ; 9141699
Glycine betaine = turgor-restoring, protein-stabilizing compatible solute	Reviews / physiology	PMID 18062278 ; 16645311

6. Supported and Refuted Hypotheses

Supported - H1: *betA* encodes an FAD-dependent, membrane-associated choline dehydrogenase whose primary reaction is choline → betaine aldehyde. **Strongly supported** (PMID 39510372). PMID 3065456

- H2: BetA is one enzyme of a coordinately regulated osmotic-stress operon (BetT import → BetA → BetB), producing the compatible solute glycine betaine. **Supported** (PMID 8626294 PMID 3065456). PMID 20923416

- H3: The enzyme is respiration-/oxygen-coupled (non-NAD acceptor; ArcA-repressed anaerobically). **Supported** (PMID 8626294). PMID 39510372

Refuted / Refined - H4 (refined): BetA is a fully bifunctional choline + betaine aldehyde dehydrogenase in *E. coli*. **Not supported for the purified *E. coli* enzyme** — pure CHDH did not oxidize betaine aldehyde onward (P 39510372); the second step is principally BetB's role. The EC 1.2.1.8 annotation reflects weak/heterologous activity only.

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7. Limitations and Future Directions

- No experimental 3-D structure of *E. coli* BetA is yet available; the catalytic His and oligomer interface are inferred from the 2024 biochemical study and GMC-family homology. A cryo-EM structure of the native trimer would clarify the active site and membrane-binding surface.
- The identity of the physiological electron acceptor (ubiquinone vs. another respiratory carrier) is inferred from EC class and ArcA regulation rather than directly demonstrated for *E. coli* BetA.
- The quantitative in vivo contribution of BetA's intrinsic betaine-aldehyde-oxidizing activity versus BetB remains to be dissected in *betB*-null *E. coli*.

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8. Conclusion

BetA (P17444) is the FAD-dependent, peripheral-inner-membrane choline dehydrogenase of *E. coli* K-12 that catalyzes the committed, aerobic oxidation of choline to betaine aldehyde — the first step of glycine betaine (osmoprotectant) biosynthesis — acting within the osmotically and choline-regulated *betIBA/betT* operon, downstream of the BetT choline transporter and upstream of the BetB betaine aldehyde dehydrogenase.
